

Models of E-Governance

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E-Governance (Electronic Governance) refers to the use of **Information and Communication Technology (ICT)** by the government to provide **efficient, effective, and transparent services** to citizens, businesses, voluntary organizations, and other government agencies.

It focuses on improving communication, access to information, and overall quality of public service delivery.

A **model** in E-Governance is a **framework or representation** that shows how digital governance functions. It acts as a **reference** that can be studied or replicated by others to achieve similar results.

Models of Digital Governance

As new technologies and governance challenges emerge, E-Governance models continue to evolve. Developing countries, including Nepal and India, are experimenting with different models to find those most suitable for their administrative structures and social needs.

Even simple models have significantly changed how information is distributed and used in society.

Some widely recognized **models of E-Governance** include:

1. **Broadcasting / Wider Dissemination Model**
2. **Critical Flow Model**
3. **Comparative Analysis Model**
4. **Mobilization and Lobbying Model**
5. **Interactive-Service Model**

1. Broadcasting / Wider Dissemination Model

This model focuses on **spreading information** that already exists within government offices (often in paper form) to a **larger audience** using ICT tools like websites, TV, radio, and online media.

It is based on **mass dissemination** of governance-related information to promote awareness among citizens about government activities, services, and their benefits.

Example

When the government **announces new rules or regulations**, it organizes press conferences, publishes them online, and broadcasts the updates through **television, radio, and official web portals** so that citizens can easily access them.

Applications

- Publishing **laws, rules, and regulations** online.
- Making available **contact details** (names, emails, and phone numbers) of government officials.
- Providing **government plans, budgets, expenditures, and performance reports** online.
- Uploading **key legal decisions** or **environmental rulings** that have public value and future importance.

Projects Based on This Model

Nepal:

- **Department of Information Technology (DoIT)** – <https://doit.gov.np>
Aims to automate government services through IT to ensure good governance, reduce poverty, and promote national development.

Brazil:

- **Official National E-Government Website** – <https://www.brasil.gov.br>

Provides comprehensive government information and integrated citizen services.

India:

- **Directory of Government of India Websites** – <http://goi-directory.nic.in/ministry.html>
Managed by the **National Informatics Centre (NIC)**, it shares information about ministries, projects, schemes, and Indian laws.

Global:

- **Earth Negotiations Bulletin** – <http://www.iisd.ca/voltoc.html>
Offers global updates on environmental negotiations and decisions, helping developing countries stay informed about international policies.

Evaluation

- Acts as the **foundation** for advanced digital governance models.
- Encourages **citizen participation** and **information transparency**.
- Becomes ineffective if **free flow of information** is restricted or if data shared is **biased or censored**.
- Requires **collaboration between government and civil society** to ensure openness and trust.

2. Critical Flow Model

The **Critical Flow Model** focuses on **sharing important or sensitive information** (of *critical value*) with the **right people or groups** using ICT and digital platforms.

The goal is to make crucial information—such as corruption cases, human rights violations, or misuse of power—**accessible to those who can take action** (like citizens, NGOs, media, or watchdog organizations).

It uses the **speed and reach of the internet** to remove barriers of distance and time, allowing important information to spread quickly and widely.

Applications:

- Publishing **corruption-related data** about ministries or officials online.
- Making **research and investigation reports** publicly available.
- Sharing **human rights violation cases** or **environmental reports** with NGOs and citizens.

Examples:

- **India – Central Vigilance Commission (CVC):**
The website cvc.nic.in publishes details of government officials charged with corruption. Citizens can also report cases online.
- **Bangladesh – Human Rights Portal (Banglarights.net):**
This portal spreads information about human rights issues, especially concerning women and children.
- **Global – Transparency International:**
The Transparency International website publishes corruption reports from around the world to raise awareness and accountability.

Evaluation:

- This model **exposes weaknesses** in governance and promotes accountability.
- It helps **citizens and organizations act** against corruption and injustice.
- However, it **fails in countries where governments censor information**, discourage public debate, or restrict media freedom.

Example Situation:

Imagine a government ministry involved in illegal land deals.

Using the **Critical Flow Model**, data about those deals is published online, accessible to journalists,

watchdog groups, and citizens.
This forces transparency and creates pressure for reform.

3. Comparative Analysis Model

The **Comparative Analysis Model** uses ICT to **compare cases of good and bad governance** so people can **learn and improve**.

By comparing performance data, policies, or officials, it helps identify what works best and what needs reform.

This model relies on the **power of digital databases and online tools** to analyze, store, and share information instantly across countries and regions.

Applications:

- Compare **past and current government policies** to measure progress.
- Evaluate **the performance of ministries or officials** using public data.
- Help citizens and decision-makers **learn from success and failure cases**.
- Support **data-driven policy-making** at all levels.

Examples:

- **India – Green Ratings Project:**
Evaluates the environmental performance of industries. Poor performers are encouraged to improve, and results are shared online for transparency.
- **Philippines – Vote.ph:**
Provides public access to records of government officials and election candidates, allowing citizens to compare their performance and make informed voting decisions.
- **Global – UNDP Human Development Indicators:**
Compares literacy, health, and income levels among countries to rank and evaluate human development progress.

Evaluation:

- Enables **learning from global and local experiences**.
- Encourages **accountability and competition** among institutions.
- Helps **citizens and media track performance** of government agencies.
- However, it **depends on data availability and citizen awareness**—without active civil society or interest, the model loses impact.

Example Situation:

Suppose two cities run different digital waste-management programs.

By comparing the results—like cost, cleanliness, and citizen feedback—through online data portals, the government can identify the **better approach** and replicate it in other regions.

4. Mobilization and Lobbying Model

This model promotes **two-way or multi-way communication** between the government and the public. It collects people's opinions (through online forums or discussions), uses them to influence policies, and strengthens real-world actions with digital participation. It focuses on **mobilizing citizens** to express their views, build momentum, and create pressure for transparent and democratic decision-making.

Applications:

- Encouraging **public debates** on important topics before major conferences or agreements.
- Forming **pressure groups** to make policymakers aware of public concerns.
- Giving **suppressed or marginalized groups** a voice in decision-making.

- Promoting **wider participation** in national or global discussions.
- Building **global expertise** when local data or insights are missing.

Examples:

- **India: Panchayats.org** – A discussion platform for professionals to share and debate local development and resource management issues.
- **Global: Independent Media Centre (Indymedia.org)** – Created during the 1999 WTO protests in Seattle to give grassroots coverage and public opinions on global trade policies.

Evaluation:

This model has grown since the WTO debates and helps expand citizen participation in policymaking. It also holds governments accountable by making them aware that their actions are being watched and discussed publicly.

5. Interactive-Service Model (Government-to–Citizen-to–Government / G2C2G)

This model also involves **two-way communication**, where citizens and the government directly interact through digital platforms. The government provides online services (like tax payment, bill submission, document access), and citizens respond or give feedback, strengthening the relationship both ways.

Applications:

- Creating **interactive communication channels** with key policymakers.
- Conducting **online voting or electronic ballots** for officials.
- Hosting **public discussions** before passing laws or policies.
- Promoting **decentralized governance** through digital tools.
- Using **video conferencing and online consultations** between citizens and government representatives.

Examples:

- **Philippines: Customs Reform Project** – Developed an online system for import clearance, duty payment, and shipment release, reducing corruption and delay.
- **India: Gyandoot Project** – A rural cybercafé network providing local people with digital services like access to land records, maps, and public notices.

Evaluation:

This model is widely used in developed nations and is considered highly interactive. However, it relies heavily on advanced ICT infrastructure, making it difficult to implement effectively in developing countries with limited internet access.

Evolution in E-Governance and Maturity Models

E-Governance is an evolutionary path whose effective implementation requires a full understanding of its various elements while staying focused on overall objectives. The e-governance journey encounters several milestones that need to be identified.

The **E-Governance Maturity Model (EMM version 1.0)** proposes five levels of maturity depending on effectiveness.

Level 1: One-way communication / Simple website / Information

- **Description:** This is the **initial stage** of e-Government. The government provides information online, but citizens cannot interact with it.
- **How it works:** Static websites are used to post information, like laws, rules, notices, or forms. Citizens can only **read or download** documents.
- **Example:** The government publishes a webpage listing **driving license rules and requirements**.

Citizens can view what documents are needed, fees, and deadlines, but they cannot submit forms online or make payments.

Level 2: Two-way interaction / Online Government / Interaction

- **Description:** At this level, **citizens can interact with the government online**. Web applications allow them to fill forms, submit documents, or make inquiries.
- **How it works:** Dynamic websites or portals enable submissions and feedback.
- **Example:** Citizens can **apply for a driving license or PAN card online**, fill the application form, upload required documents, and make payment through the portal. The system receives the request and may send confirmation emails or notifications.

Level 3: Centralized system / Integrated Government / Transaction

- **Description:** This level introduces **centralized management and full integration**. Multiple services and departments are connected to a single portal for efficiency.
- **How it works:** Citizens can perform complete transactions online, and the government can process data centrally. Time and cost are reduced, and processes become more transparent.
- **Example:** A citizen **renews a driving license online**. The system checks previous records, verifies payment, and schedules a slot for a driving test if required—all managed centrally without needing to visit multiple offices.

Level 4: Transformed Government / Transformation

- **Description:** This is the **most advanced stage**. Processes are **automated**, integrated, and require minimal human intervention. Complex workflows are handled digitally, and the system may even generate reports or refunds automatically.
- **How it works:** Intelligent systems process applications, approvals, and notifications automatically. E-services are seamless and user-friendly.
- **Example:** **PAN card renewal or tax filing system:** the system auto-verifies documents, processes payments, sends e-confirmation, and even auto-generates refunds or updates the citizen's profile without manual handling.

A Maturity Model

- A maturity model is a method for judging **the maturity of organizational processes**.
- Helps identify weaknesses and provides solutions to improve systems and processes.
- Measures methods and processes against **external benchmarks**.

Benefits of maturity assessment:

- Compare maturity levels across ministries/departments.
- Know the organization's maturity level with precise recommendations.
- Provides independently held benchmarks.

Five Maturity Levels of E-Governance

Level 1: Closed

- The organization does not use computers or the internet.
- Reasons: Lack of knowledge, money, or awareness.
- Everything is done manually, and information is not shared.
- **Example:** A small government office where people have to visit in person to apply for a driving license or PAN card.

Level 2: Initial

- Some computer-based systems are used, but they are not organized or consistent.
- No full plan for e-governance.
- Computers or software may be available but not used properly.
- **Example:** An office tries online PAN card applications, but the system works only sometimes and staff still handle many things manually.

Level 3: Planned

- A clear plan is made with goals and priorities.
- Areas for e-governance are chosen carefully.
- **Example:** Planning a website where citizens can apply for driving licenses, PAN cards, and pay taxes in one place.

Level 4: Realized

- The e-governance plan is fully implemented.
- Services are delivered to both citizens and government staff efficiently.

Sub-levels:

1. **Retrospected** – Processes are checked and improved.
 2. **E-ready** – All necessary computers, software, and networks are ready.
 3. **Partially open** – Some services are online, but not everything.
 4. **Open** – All planned services are available online and work well.
- **Example:** Citizens can **apply for a driving license, renew PAN card, or pay taxes online**, check the status, and get updates.

Level 5: Institutionalized (Established)

- The e-governance system is fully established and becomes part of daily work.
- People use it regularly, and it works smoothly.
- The organization has the right leadership, infrastructure, staff, and laws to support it.
- **Example:** The government's **e-tax system** is widely used, connected with banks and other agencies, and regularly updated for new rules.

Characteristics of Maturity Levels

Level 1: Closed

- Organization is closed to e-governance.
- No plans or vision for e-governance.
- Operations are fully manual and conventional.

Level 2: Initial

- Lacks strategic thinking and direction at the top level.
- Unorganized, isolated automation efforts in some areas.
- Hardware may be accumulated but remains unutilized or underutilized.

Level 3: Planned

- E-governance becomes part of the organization's agenda.
- Strategic thinking and leadership guide initiatives.
- Clear understanding of internal and external user needs.

Level 4: Realized

- **Retrospected:**
 - Business processes aligned with vision and overall e-governance objectives.
 - Awareness among stakeholders and users about e-governance.
- **E-ready:**
 - Sound ICT and institutional infrastructure in place.
 - Users motivated to use e-governance services.
- **Partially Open:**
 - Focused mainly on internal processes; information exchange confined within the organization.
 - G2E (Government to Employee) exists; G2C, G2G, G2B not yet established.
- **Open:**
 - Fully integrated system allowing smooth information exchange internally and externally.
 - G2E, G2C, G2G, G2B fully functional.

Level 5: Institutionalized

- E-governance becomes a natural part of organizational operations.
- Fully paperless system.
- Services are effectively utilized and accepted by stakeholders and customers.

Towards Good Governance through E-Governance Models

E-Governance is key to **good governance**, especially in developing countries, by:

- Minimizing corruption.

- Providing efficient, effective, and quality services to citizens.

Over the last decades, lack of transparency and various scams highlighted the need for:

"A government that is responsive to the needs of people, with clearly defined, time-bound accountability and decisions readily available to the public."

Principles for Good Governance through E-Governance

- **Accountability:**
 - Define personnel and stakeholders responsible for service delivery.
 - Ensure clear responsibility for tasks and decisions.
- **Daily Administration Problems:**
 - Citizens know whom to question and expected timelines for services.
- **Interdepartmental Problems:**
 - Departments' jurisdictions are clearly defined to avoid delays.
- **Policy-Level Issues:**
 - Complex, long-term issues addressed through careful analysis.
- **Transparency:**
 - Automated systems ensure highly transparent transactions.
- **Responsive:**
 - Systems must meet the requirements of different population sections efficiently.